

❖ Section 7. Markov Chain Monte Carlo

Instructor: He Wang
Department of Mathematics
Northeastern University

- Foundation of MCMC
- Basic Idea and examples of MCMC
- Basic Metropolis algorithm
- Metropolis-Hastings algorithm
- Gibbs Sampling

Basic Idea:

MCMC techniques have become pervasive and essential in most aspects of Data Analysis and Applied Probability. By using powerful computing resources the MCMC technique allows answers to be found for highly complicated problems involving statistical optimization.

[The Markov chain Monte Carlo revolution](#)--[Persi Diaconis](#)

<https://www.ams.org/journals/bull/2009-46-02/S0273-0979-08-01238-X/>

➤ Markov Chain Monte Carlo

- The MCMC sampling strategy is to construct an irreducible, aperiodic Markov Chain for which the stationary distribution equals the target distribution.
- If we run the chain long enough, it will give us samples from the target distribution, and then can examine their mean, variance and so on.
- Popular MCMC approaches: Metropolis-Hastings Algorithm, Gibbs Sampling
- Popular application: estimate posterior distribution in Bayesian Statistics

Foundation of MCMC

- Recall that we use Monte Carlo method to estimate

$$E[h(X)] := \int h(x)p(x)dx \quad \text{or} \quad E[h(X)] := \sum_{j=1}^{\infty} h(x_j)P(X = x_j)$$

by generating **iid** random values $X_1, X_2, \dots, X_n$ with pmf $P(X = x_j)$, for $j \geq 1$.

$$\frac{1}{n} \sum_{i=1}^n h(X_i) \longrightarrow E[h(X)] \text{ as } n \rightarrow \infty$$

- Recall ergodic theorem that

Theorem. If (non-iid) $X_1, X_2, \dots$ are realizations from an **irreducible, aperiodic, and positive recurrent** Markov chain with stationary distribution π , then, for any bounded function $h: S \rightarrow \mathbb{R}$,

$$\frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{a.s.}} E_{\pi}[h(X)] = \sum_{j=1}^{\infty} h(j)\pi_j \quad \text{as } n \rightarrow \infty$$

The basic idea of MCMC

- Let $\Omega = \{1, 2, \dots, K\}$ be a finite state space.
- Let π be a **target** probability distribution on Ω .

Goal: generate random samples from π , meaning to generate elements of Ω which are chosen randomly with probability distribution π .

The MCMC method proceeds by constructing an ergodic Markov Chain on Ω for which π is the stationary distribution.

By running the Markov Chain, we generate a sequence of states

$$X_1, X_2, \dots,$$

whose distribution is guaranteed to converge to π .

So, **for any** $\epsilon > 0$, **there is** N sufficiently large such that the sequence

$$X_N, X_{N+1}, \dots$$

has distribution close to π , in the sense that, for all $k \geq 0$,

$$\sum_{j \in \Omega} |P(X_{N+k} = j) - \pi(j)| \leq \epsilon$$

Long-run fraction of time

- (X_n) is an **ergodic Markov chain** with stationary distribution π .
- $A \subset \Omega$ is a **nonempty** subset of the state space Ω .
- Define

$$\pi_A := \sum_{j \in A} \pi_j .$$

Encode “being in A ” by an **indicator function**

$$I_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then for a sample path $X_0, X_1, \dots$,

$$\sum_{k=0}^{n-1} I_A(X_k)$$

is **exactly the number of visits to A** in the first n steps.

Apply the ergodic theorem for $I_A(x)$, with probability 1,

$$\frac{1}{n} \sum_{k=0}^{n-1} I_A(X_k) \rightarrow \mathbb{E}_\pi[I_A(X)].$$

The stationary expectation $X \sim \pi$, the **long-run fraction of time** that an ergodic Markov chain spends in the set A .

$$\mathbb{E}_\pi[I_A(X)] = \sum_{j \in \Omega} I_A(j) \pi_j = \sum_{j \in A} \pi_j = \pi_A$$

Further more,

$$\mathbb{E}(\text{time between visits to } A) = \frac{1}{\pi_A},$$

Example. Bob's daily lunch choices at the cafeteria are described by a Markov chain with transition matrix P

	Yogurt	Salad	Hamburger	Pizza
Yogurt	0	0	1/2	1/2
Salad	1/4	1/4	1/4	1/4
Hamburger	1/4	0	1/4	1/2
Pizza	1/4	0	1/4	1/2

Yogurt costs \$3.00, hamburgers cost \$7.00, and salad and pizza cost \$4.00 each. Over the long term, how much, on average, does Bob spend for lunch?

Let the state space be $\{Y, S, H, P\}$. Define **reward/cost** function

$$r(x) = \begin{cases} 3, & x = Y, \\ 4, & x \in \{S, P\}, \\ 7, & x = H. \end{cases}$$

So the total cost over n days is $\sum_{k=0}^{n-1} r(X_k)$, and the **average daily cost** is

$$\frac{1}{n} \sum_{k=0}^{n-1} r(X_k).$$

Use the ergodic theorem

$$\frac{1}{n} \sum_{k=0}^{n-1} r(X_k) \xrightarrow[n \rightarrow \infty]{a.s.} \mathbb{E}_{\pi}[r(X)] = \sum_x r(x) \pi_x.$$

Stationary distribution

$$\pi_Y = \frac{7}{65}, \pi_S = \frac{2}{13}, \pi_H = \frac{18}{65}, \pi_P = \frac{6}{13}.$$

So

$$\sum_x r(x) \pi_x = 3 \cdot \frac{7}{65} + 4 \left(\frac{2}{13} + \frac{6}{13} \right) + 7 \cdot \frac{18}{65} = \frac{307}{65} \approx 4.72.$$

This example is **exactly** the “set A ” idea, but with a **reward (cost) function** instead of an indicator.

❑ MCMC-Binary Sequences with No Adjacent 1s

A binary sequence $x = (x_1, \dots, x_m) \in \{0,1\}^m$, is called **good** if it has no adjacent 1s:

For example ($m = 4$), there are $2^4 = 16$ binary sequences total. The 8 good ones are:

0000, 1000, 0100, 0010, 0001, 1010, 1001, 0101

Let G_m denote the set of all good sequences of length m .

Define the reward function

$$r(x) = \sum_{i=1}^m x_i = \text{number of 1s in } x$$

All good sequences are equally likely, the expected number of 1s is

$$\mu = E[r(X)] \qquad X \sim \text{Uniform}(G_m)$$

For example, ($m = 4$), $\mu = \frac{1}{8}(0 + 1 + 1 + 1 + 1 + 2 + 2 + 2) = 1.25$

A natural idea is **rejection sampling**:

- Generate a uniformly random binary sequence of length m
- Accept it if it is good; otherwise reject and repeat

For $m = 100$: Combinatorics becomes difficult.

Let $c_m = |G_m|$, the numbers c_m satisfy the Fibonacci recurrence

$$c_m = c_{m-1} + c_{m-2} \text{ with } c_0 = 1, c_1 = 2$$

Then the acceptance probability is

$$P(\text{accept}) = \frac{c_m}{2^m} \approx \frac{10^{21}}{10^{30}} \approx \frac{1}{10^9}$$

MCMC Strategy

Instead of sampling independently, we construct a **Markov chain** whose:

- State space is G_m
- Stationary distribution is **uniform on** G_m

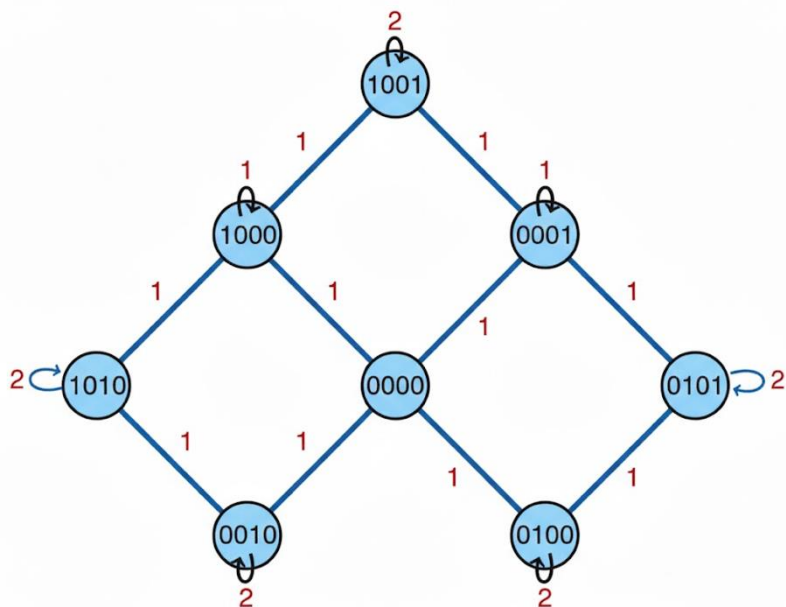
Given a current good sequence x :

1. Choose an index $i \in \{1, \dots, m\}$ uniformly at random
2. If $x_i = 1$, flip it to 0 (the result is always good)
3. If $x_i = 0$:
 - Flip it to 1 **only if** the resulting sequence remains good
 - Otherwise, stay at the current state

This defines a **lazy random walk** on the space of good sequences.

The Markov chain can be viewed as a random walk on a **weighted graph**:

- Vertices: elements of G_m
- Two vertices are connected if they differ in exactly one bit
- Each valid flip has weight 1
- Each vertex has a self-loop whose weight makes the total outgoing weight equal to m



$$P = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & \frac{1}{4} \\ \frac{1}{4} & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & 0 & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & \frac{1}{4} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{pmatrix}.$$

Estimating the Expectation via the Ergodic Theorem

Let $(X_t)_{t \geq 0}$ be the Markov chain and define

$$r(X_t) = \text{number of 1s at time } t.$$

By the **strong law of large numbers for Markov chains**,

$$\frac{1}{n} \sum_{t=1}^n r(X_t) \xrightarrow[n \rightarrow \infty]{a.s.} \mathbb{E}_{\pi}[r(X)],$$

where π is the stationary distribution (uniform on G_m). Thus,

$$\hat{\mu}_n = \frac{1}{n} \sum_{t=1}^n r(X_t)$$

is a **consistent MCMC estimator** of μ .

Numerical Calculation ($m = 100$)

https://colab.research.google.com/drive/1bQP3zK90WzvMSZbfBC4Uo4gfoxs_YOQsc#scrollTo=f1112d3f

❖ The Basic Metropolis Algorithm

- Let $\Omega = \{1, 2, \dots, K\}$ be a finite state space,
- Let $\pi = (\pi(i))_{i \in \Omega}$ be a target probability distribution satisfying

$$\pi(i) > 0 \text{ for all } i \in \Omega.$$

Metropolis Proposal Chain

Let $Q = (q_{ij})$ be a **known proposal transition matrix** on Ω with the following properties:

- **Symmetric:** $q_{ij} = q_{ji}$ for all i, j
- **Irreducible**
- **Aperiodic**

Because Q is symmetric, its stationary distribution is the **uniform distribution** on Ω . (The Metropolis–Hastings algorithm extends this construction to non-symmetric Q .)

$$Q = \begin{bmatrix} q_{11} & q_{12} & \cdots & q_{1K} \\ q_{21} & q_{22} & & q_{2K} \\ \vdots & & \ddots & \vdots \\ q_{K1} & q_{K2} & \cdots & q_{KK} \end{bmatrix} = \begin{bmatrix} -q_{1j} - \\ -q_{2j} - \\ \vdots \\ -q_{1j} - \end{bmatrix}$$

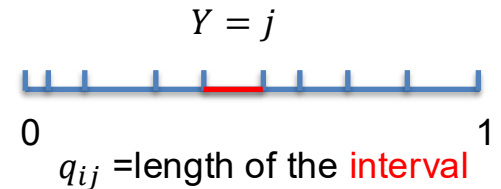
Goal of MCMC: Transform the proposal chain Q into a new Markov chain $P = (p_{ij})$ whose stationary distribution is the desired target distribution π .

The algorithm constructs a sequence $X_0, X_1, X_2, \dots$, where X_0 is any chosen initial state, and X_{k+1} **is constructed from X_k** using the following steps.

Metropolis Algorithm

1. Randomly select $Y \in \Omega$ according to the probability distribution Q :

$$P(Y = j | X_k = i) = q_{ij}$$



2. Define the **acceptance** probability:

$$\alpha := \alpha(Y | X_k = i) := \min \left\{ 1, \frac{\pi(Y)}{\pi(X_k)} \right\} = \min \left\{ 1, \frac{\pi(j)}{\pi(i)} \right\}, \quad \text{if } Y = j \text{ and } X_k = i$$

3. Define

$$X_{k+1} = \begin{cases} Y & \text{with probability } \alpha \\ X_k & \text{with probability } 1 - \alpha \end{cases}$$

- For Step 3, more concretely, we flip a weighted coin with probability of Heads equal to α :
 - if it comes up Head we set $X_{k+1} = Y$,
 - if Tail we set $X_{k+1} = X_k$).
- It can also be achieved by the following:
 - Generate a random $u \sim \text{Uniform}(0,1)$
 - Define
$$X_{k+1} = \begin{cases} Y & \text{if } u < \alpha \\ X_k & \text{if } u \geq \alpha \end{cases}$$

Transition matrix

- This protocol defines the **off-diagonal** entries of the transition matrix

$P = (p_{ij})$ for the new Markov chain, namely

$$p_{ij} := P(X_{k+1} = j \mid X_k = i) = q_{ij} \alpha = q_{ij} \min \left\{ 1, \frac{\pi(j)}{\pi(i)} \right\} \text{ for } i \neq j$$

We then define the **diagonal** entries, in order to make the matrix stochastic:

$$p_{ii} := 1 - \sum_{j \neq i} p_{ij}$$

Time Reversible Markov chain:

The sequence $X_0, X_1, X_2, \dots$, defined by the Metropolis algorithm is time reversible.

Theorem. For all $i, j \in \Omega$, we have the following

$$\pi(i)p_{ij} = \pi(j)p_{ji}$$

Proof: It is sufficient to assume that $i \neq j$, in which case,

$$\pi(i)p_{ij} = \pi(i)q_{ij} \min \left\{ 1, \frac{\pi(j)}{\pi(i)} \right\}$$

$$= q_{ij} \min\{\pi(i), \pi(j)\}$$

$$= q_{ji} \pi(j) \min \left\{ 1, \frac{\pi(i)}{\pi(j)} \right\}$$

$$= \pi(j)p_{ji}$$

since Q is symmetric,
 $q_{ij} = q_{ji}$.

Remarks: This theorem shows that the new Markov chain is time **reversible**, and therefore it follows that π is its stationary distribution. So $\{X_k\}$ is the promised Markov chain that converges to π . Note also that the computation of p_{ij} requires only knowledge of the ratio $\frac{\pi(j)}{\pi(i)}$.

We will see that there are many applications where these ratios can be easily computed, while the probabilities $\pi(i)$ are themselves difficult to find, because the overall normalization is hard to compute. One important case is Bayesian inference.

Example: Sampling from a multinomial distribution

Suppose that X is discrete with $\text{Range}(X) = \{1, 2, \dots, K\}$, and pdf

$$P(X = k) = q_k, \quad k = 1, \dots, K$$

We will use MCMC to estimate the **expected value** $E[X^2]$.

So, our stationary pdf for the Markov chain is $\pi = (q_1, \dots, q_K)$.

We choose the reference transition to be **uniform** on the state space, so

$$q_{ij} = \frac{1}{K} \quad \text{for all } i, j = 1, \dots, K$$

Then the transition probability for MCMC is

$$p_{ij} = \begin{cases} \frac{1}{K} \min\left(1, \frac{q_j}{q_i}\right) & \text{for all } i \neq j \\ 1 - \sum_{k \neq i} p_{ik} & \text{for } j = i \end{cases}$$

So, starting with an initial state X_0 , we simulate a sequence $X_1, X_2, \dots$ using this transition matrix.

We then estimate the mixing time of the chain - this is the number of steps s such that X_n and X_{n+s} are approximately independent. Then we estimate the expected value as

$$E[X^2] \approx \frac{1}{N} \sum_{n=1}^N X_{ns}^2$$

Let's compute a concrete example ([Colab](#)).

<https://colab.research.google.com/drive/1Q7ivKVM9qrIlhwYWI6BQ1sJbdLhMveXA?usp=sharing>

Example: Sampling from a Uniform distribution

Many interesting problems require computing the **average of functions** over complicated discrete sets. Such averages may be estimated by drawing independent samples from the set (IID samples) and computing the empirical average of the function over these samples.

Namely, let Ω be a finite discrete set, and $f: \Omega \rightarrow \mathbb{R}$.

Suppose we want to estimate the average

$$E[f] = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} f(\omega)$$

Let $X_1, \dots, X_k$ be drawn randomly and uniformly from Ω , and define

$$\bar{f}_k = \frac{1}{k} \sum_{i=1}^k f(X_i)$$

Then,

$$\lim_{k \rightarrow \infty} \bar{f}_k = E[f]$$

where the expectation on the right side is taken with respect to the uniform distribution on Ω , and the convergence is almost sure.

Example: knapsack problem

The knapsack problem is a problem in combinatorial **optimization**: Given a set of items, each with a mass and a value, determine the number of each item to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible. Knapsack problem is NP complete.



One-dimensional knapsack problem:

Which books should be chosen to maximize the amount of money while still keeping the overall weight under or equal to 15 kg?

Example: knapsack problem

Let $z = (z_1, \dots, z_m)$.

Each $z_i \in \{0,1\}$ and so $z \in \{0,1\}^m$

Let $\{w_i\}$ be positive scalars, and $b > 0$.

Define the discrete set

$$\Omega := \left\{ z \in \{0,1\}^m \mid \sum_{j=1}^m w_j z_j \leq b \right\}$$

Assuming that $\sum_j w_j > b$, it follows that Ω is a proper (and complicated) subset of the hypercube $\{0,1\}^m$.

Goal 1: We want to sample randomly and uniformly from Ω .

Goal 2: Maximize $\sum_{j=1}^m v_j z_j$.

By using the MCMC method to construct an appropriate Markov chain.

Since the target distribution is uniform, the Markov chain should be symmetric.

The update rule is computed as follows:

Given that $X_k = z = (z_1, \dots, z_m)$

Let $J \in \{1, \dots, m\}$ be a uniform random variable (independent of all other r.v.'s) and define

$$Y = (z_1, \dots, 1 - z_J, \dots, z_m)$$

In other words, we flip the J -th bit to get the new point. Then we define

$$X_{k+1} = \begin{cases} Y & \text{if } Y \in \Omega \\ X_k & \text{if } Y \notin \Omega \end{cases}$$

The chain is clearly **irreducible** since it is possible to reach the state $(0, \dots, 0)$ starting from any state.

Also, since Ω is a proper subset of the hypercube, there is a state z and a choice of J such that $Y \notin \Omega$ and hence $X_{k+1} = X_k$; therefore the chain is **aperiodic**.

Thus, the chain is **ergodic**, and so **converges** to its unique stationary distribution.

Finally, the following computation shows that the chain is **symmetric**:

Let $w, z \in \Omega$ with $w \neq z$ such that there is $j \in \{1, \dots, m\}$ with

$$w_i = z_i, \quad \text{for } i \neq j$$

$$w_j = 1 - z_j$$

Hence

$$P(X_{k+1} = w | X_k = z) = P(J = j) = \frac{1}{m}$$

Similarly,

$$P(X_{k+1} = z | X_k = w) = P(J = j) = \frac{1}{m}$$

So, the transition matrix is symmetric for this pair.

Furthermore, the transition matrix between states $w \neq z \in \Omega$ is nonzero if and only if $w_j = 1 - z_j$ for some j .

So, this shows that the transition matrix is symmetric, and hence that the stationary distribution is uniform.

We can now use this Markov chain to estimate the average of a function on Ω .

Consider for example the function

$$f(z) = e^{\beta \sum_j v_j z_j}$$

where $\beta > 0$ and $\{v_j\}$ is a fixed vector.

The average is

$$E[f] = \frac{1}{|\Omega|} \sum_{z \in \Omega} e^{\beta \sum_j v_j z_j}$$

(This is similar to the partition function encountered in statistical mechanics, where β is the inverse temperature.)

By running the Markov chain, we generate samples X_k drawn uniformly from Ω .

Let

$$Y_n = X_{N_0 + nN_1} \quad \text{for } n = 1, 2, \dots \text{ and } N_0, N_1 \in \mathbb{Z}$$

Our Estimate of $E[f]$ is

$$\hat{f}_K = \frac{1}{K} \sum_{n=1}^K f(Y_n)$$

By choosing N_0 large we ensure that the marginal distribution of each Y_n is close to uniform, so

$$E[\hat{f}_K] \simeq E[f]$$

By choosing N_1 large we ensure that the different Y_n are close to being independent, and hence

$$\text{Var}[\hat{f}_K] \simeq \frac{1}{K} \text{Var}[f]$$

Example: Maximizing a function

Continuing with the knapsack problem, we may wish to find the maximum value of f over Ω .

To do this, we construct a new MCMC algorithm whose stationary distribution is

$$\pi_{\beta}(z) = \frac{1}{E[f]} f(z) = \frac{1}{E[f]} e^{\beta \sum_j v_j z_j}$$

For large β this distribution is tightly concentrated around the value of z where f attains its maximum.

Thus, by sampling from π with large β we will with high probability generate a value of z which is close to the maximal value.

We modify the MCMC construction from the previous section as follows:

Given that $X_k = z = (z_1, \dots, z_m)$.

Let $J \in \{1, \dots, m\}$ be a uniform random variable (independent of all other r.v.'s).

Define

$$Y = (z_1, \dots, 1 - z_J, \dots, z_m)$$

If $Y \notin \Omega$, then $X_{k+1} = X_k$

If $Y \in \Omega$, then let $\alpha = \min \left\{ 1, \frac{\pi_\beta(Y)}{\pi_\beta(X_k)} \right\} = \min \left\{ 1, e^{\beta v_J(1-2z_J)} \right\}$

and define

$$X_{k+1} = \begin{cases} Y & \text{with probability } \alpha \\ X_k & \text{with probability } 1 - \alpha \end{cases}$$

Notice that the rejection probability increases as β increases, and this slows down the convergence of the algorithm. Simulated annealing is the method of slowly increasing β as the chain progresses, thereby focusing in on the region of high probability where we are most interested.

Example: Bayesian Inference

Let θ be a parameter in a family of probability distributions. Given a collection of data $\mathcal{D}$, our goal is to find the best estimate of θ using this data.

The Bayesian method requires the choice of an initial or prior distribution for θ , let us denote this by P_0 . Then the Bayesian formula for the posterior distribution P_1 is

$$P_1(\theta|\mathcal{D}) = \frac{P(\mathcal{D}|\theta)P_0(\theta)}{\int P(\mathcal{D}|\theta')P_0(\theta')d\theta'}$$

Often the likelihood function and the prior can be easily evaluated but the evidence is hard to compute. This is a situation where MCMC can be very helpful.

Example. Suppose that at a party each entering guest is given a card with a number on it. The cards are numbered 1,2,3, ... and they are handed out in sequence to entering guests. At the end of the evening many guests have left, but the host cannot remember how many cards were handed out, although she knows that the number was between 100 and 200. There are 20 people left in the room and she records the numbers on their cards. Given this information she tries to estimate the total number of guests. How should she do this?

Answer by Bayesian inference.

Let M be the (unknown) number of guests.

Let $x_1, \dots, x_k$ be the numbers on the cards of the remaining guests ($k = 20$)

So as random variables the data is $D = (X_1, \dots, X_k)$ and the parameter is M .

The prior for M is uniform on the interval $(n_1, n_2]$ (in this case $n_1 = 100$ and $n_2 = 200$). The likelihood is

$$P(D|M = m) = \prod_{i=1}^k P(X_i = x_i|M = m) = \prod_{i=1}^k \frac{1}{m} \theta(x_i \leq m)$$

Here $\theta(A)$ is the indicator function for the event A :

$$\theta(A) = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{if } A \text{ is false} \end{cases}$$

Hence

$$P(D|M = m) = \frac{1}{m^k} \theta(\max_i x_i \leq m)$$

The **prior** for M is

$$P_0(M = m) = \frac{1}{n_2 - n_1} \theta(n_1 < m \leq n_2)$$

So, the **evidence** is

$$Z = \sum_m P(D|M = m)P_0(M = m) = \frac{1}{n_2 - n_1} \sum_{m=\max\{\max x_i, n_1+1\}}^{n_2} m^{-k}$$

Hence the **posterior** probability distribution for M is

$$P_1(M = m|D) = \frac{1}{Z} \frac{1}{n_2 - n_1} m^{-k} \theta(\max\{\max x_i, n_1 + 1\} \leq m \leq n_2)$$

One popular statistic is the maximum likelihood estimator $\hat{M}_{MLE}$. This is the value of m which maximizes the posterior probability, that is the value which makes it most likely that the given data would have been observed.

Clearly the MLE here is the maximum of the observed values, that is

$$\hat{M}_{MLE} = \max x_i$$

However we might also be interested in the mean of the posterior, that is

$$\bar{M} = \sum_m m P_1(M = m | D)$$

In this case we must evaluate the evidence Z , which of course is difficult to do analytically. So, we use MCMC to simulate the distribution. Define

$$a = \max\{\max x_i, n_1 + 1\}, \quad b = n_2$$

The state space is $\Omega\{a, a + 1, \dots, b\}$ and we use a nearest neighbor model for our basic chain. Then the transition matrix is

$$q_{i,i+1} = q_{i,i-1} = \frac{1}{2} \quad \text{for } a \leq i \leq b$$

$$q_{a,a} = q_{b,b} = \frac{1}{2}$$

This chain is aperiodic and irreducible. The MCMC transition matrix is then

$$p_{i,i+1} = \frac{1}{2} \left(\frac{i+1}{i} \right)^{-k} \quad \text{for } i = a, a + 1, \dots, b - 1$$

$$p_{i,i-1} = \frac{1}{2} \quad \text{for } i = a + 1, \dots, b$$

So the MCMC chain is a random walk on the integers with bias to the left.

As an example, suppose that $n_1 = 100, n_2 = 200, k = 20$ and $\max x_i = 140$. Then $a = 140$ and $b = 200$.

We construct the Markov chain using the transition matrix above.

So for example, $p_{150,151} = 0.4378$

We want to compute the mean of the posterior distribution, so we approximate this by

$$\bar{X} = \frac{1}{K} \sum_{i=1}^K Y_i$$

where $\{Y_i\}$ are samples from the chain. We want these to be approximately independent, and also approximately stationary.

So we choose a random initial state X_0 , and then generate $X_1, \dots, X_N$, where N is chosen large enough so that the initial condition is forgotten.

Then we set $Y_1 = X_N$. We continue, and generate $Y_2 = X_{2N}, Y_3 = X_{3N}, \dots, Y_K = X_{KN}$.

Using $N = K = 10,000$ we find for this example that

$$\bar{X} \approx 147.2 \pm 0.2$$

❖ Metropolis-Hastings algorithm

Metropolis et al. (1953, the fifth author was Edward Teller, father of the hydrogen bomb") simulated a liquid in equilibrium with its gas phase. This is the algorithm we just learned.

Hastings (1970) generalized the Metropolis algorithm. The Metropolis–Hastings algorithm is a Markov chain Monte Carlo (MCMC) method for obtaining a sequence of random samples from a probability distribution from which direct sampling is difficult.

Most of the MCMC are either a special case or a generalization of the Metropolis–Hastings algorithm.

For a target distribution $p(x)$, we need a proposal distribution $q(x'|x)$ that is easy to sample from. The algorithm constructs a Markov Chain with stationary distribution f .

Metropolis-Hastings algorithm

The Metropolis–Hastings (MH) algorithm generalizes the Metropolis algorithm by allowing a **non-symmetric proposal distribution**.

Setup. Let

$$\Omega = \{1, 2, \dots, K\} \text{ (finite state space)}$$

and let $\pi = (\pi(i))$ be a target distribution satisfying

$$\pi(i) > 0 \text{ for all } i \in \Omega.$$

Let

$$Q = (q_{ij})$$

be a **proposal transition matrix** on Ω such that

- $q_{ij} \geq 0$, $\sum_j q_{ij} = 1$
- Q is **irreducible** and **aperiodic**
- Q is **not required to be symmetric**

Our goal is to construct a Markov chain with stationary distribution π .

Metropolis–Hastings Algorithm

Let $\{X_k\}_{k \geq 0}$ be a Markov chain with arbitrary initial state $X_0 \in \Omega$.

Given $X_k = i$, construct X_{k+1} as follows.

1. Sample a **proposal** state $Y \in \Omega$ according to

$$\mathbb{P}(Y = j \mid X_k = i) = q_{ij}.$$

2. Define the **Metropolis–Hastings acceptance probability**

$$\alpha(Y \mid X_k = i) := \min \left\{ 1, \frac{\pi(Y) q_{Yi}}{\pi(X_k) q_{iY}} \right\} = \min \left\{ 1, \frac{\pi(j) q_{ji}}{\pi(i) q_{ij}} \right\}.$$

3: **Accept or Reject.** Set

$$X_{k+1} = \begin{cases} Y, & \text{with probability } \alpha, \\ X_k, & \text{with probability } 1 - \alpha. \end{cases}$$

Equivalently:

- Draw $u \sim \text{Uniform}(0, 1)$
- If $u < \alpha$, accept the proposal; otherwise, stay at X_k

Transition Matrix of the MH Chain

For $i \neq j$, the off-diagonal transition probabilities are

$$p_{ij} = q_{ij} \min \left\{ 1, \frac{\pi(j)q_{ji}}{\pi(i)q_{ij}} \right\}.$$

The diagonal entries are defined by

$$p_{ii} = 1 - \sum_{j \neq i} p_{ij},$$

so that each row of $P = (p_{ij})$ sums to 1.

Time Reversibility and Stationarity

Theorem (Detailed Balance)

The Metropolis–Hastings chain satisfies

$$\pi(i)p_{ij} = \pi(j)p_{ji} \text{ for all } i, j \in \Omega.$$

Proof

For $i \neq j$,

$$\pi(i)p_{ij} = \pi(i)q_{ij} \min \left\{ 1, \frac{\pi(j)q_{ji}}{\pi(i)q_{ij}} \right\} = \min \{ \pi(i)q_{ij}, \pi(j)q_{ji} \}.$$

By symmetry of the minimum,

$$\min \{ \pi(i)q_{ij}, \pi(j)q_{ji} \} = \pi(j)q_{ji} \min \left\{ 1, \frac{\pi(i)q_{ij}}{\pi(j)q_{ji}} \right\} = \pi(j)p_{ji}.$$

Consequences

- The chain is **time reversible** with respect to π
- Hence π is a **stationary distribution**
- Since the chain is irreducible and aperiodic,

$$X_k \xrightarrow{d} \pi$$

The Metropolis–Hastings algorithm requires **only ratios**

$$\frac{\pi(j)}{\pi(i)},$$

So, the normalizing constant of π cancels out.

If the target distribution $\pi(x) = \frac{\tilde{\pi}(x)}{M}$ with unknown M , we can still use the Metropolis-Hastings algorithm by using $\tilde{\pi}(x)$ in the algorithm.

This is why MH is fundamental in:

- Bayesian inference
- Posterior sampling
- High-dimensional statistical models

Example: Power-law distributions

Power-law distributions are positive probability distributions of the form

$$\pi_i \propto i^s,$$

for some constant s . Power-law distributions have **fat tails** and are often used to model highly skewed data.

Consider the probability distribution $\pi = (\pi_1, \pi_2, \dots)$ on the positive integers defined by

$$\pi_i = \frac{i^{-3/2}}{\sum_{k=1}^{\infty} k^{-3/2}}, \quad i = 1, 2, \dots$$

Implement a Metropolis–Hastings algorithm to simulate from.

You never need the normalizing constant $\sum_{k \geq 1} k^{-3/2}$

Proposal (reflecting random walk on $\{1, 2, 3, \dots\}$)

If $i = 1$: propose $j = 2$ with probability 1.

If $i \geq 2$: propose $j = i - 1$ or $j = i + 1$ with probability $\frac{1}{2}$ each.

So

$$T_{ij} = \begin{cases} 1, & i = 1, j = 2, \\ \frac{1}{2}, & i \geq 2, j = i \pm 1, \\ 0, & \text{otherwise.} \end{cases}$$

Acceptance probability

Metropolis–Hastings acceptance:

$$\alpha(i, j) = \min\left\{1, \frac{\pi(j)T_{ji}}{\pi(i)T_{ij}}\right\}.$$

Because $\pi(i) \propto i^{-3/2}$, for neighbors $i \leftrightarrow i + 1$ with $i \geq 2$,

Up move $i \rightarrow i + 1$:

$$\alpha(i, i + 1) = \min\left\{1, \frac{(i + 1)^{-3/2}}{i^{-3/2}}\right\} = \left(\frac{i}{i + 1}\right)^{3/4}.$$

Down move $i + 1 \rightarrow i$: the ratio is ≥ 1 , so $\alpha(i + 1, i) = 1$.

At the boundary $1 \leftrightarrow 2$,

$$\alpha(1, 2) = \min\left\{1, \frac{\pi(2)T_{21}}{\pi(1)T_{12}}\right\} = \min\left\{1, \frac{2^{-3/2} \cdot \left(\frac{1}{2}\right)}{1 \cdot 1}\right\} = 2^{-5/2}.$$

(and $\alpha(2, 1) = 1$)

□ Multivariate by univariate sampling

Motivation

- **Rejection sampling** is effective only in **low dimensions**; its acceptance probability typically decays exponentially with dimension.
- **Direct sampling** from high-dimensional joint distributions is generally difficult.
- A powerful strategy is to **reduce multivariate sampling to a sequence of lower-dimensional (often univariate) sampling problems**.

This idea underlies **ancestral sampling** (when a factorization is available) and **Gibbs sampling** (when only conditional distributions are tractable).

Ancestral sampling

Factorization provides a recipe for data generation / sampling from target $p(x)$.

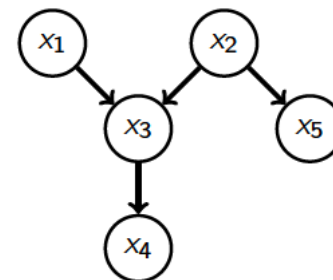
Suppose a joint distribution factorizes as :

$$p(x_1, \dots, x_5) = p(x_1)p(x_2)p(x_3|x_1, x_2)p(x_4|x_3)p(x_5|x_2)$$

Such a factorization (e.g., from a Bayesian network / DAG) provides a **direct recipe for sampling**.

We can generate samples from the joint distribution $p(x_1, \dots, x_5)$ by sampling

1. $x_1 \sim p(x_1)$
2. $x_2 \sim p(x_2)$
3. $x_3 \sim p(x_3|x_1, x_2)$
4. $x_4 \sim p(x_4|x_3)$
5. $x_5 \sim p(x_5|x_2)$



This reduces multivariate sampling to a **set of univariate (or low-dimensional) sampling problems**.

Limitation

Ancestral sampling requires:

- An explicit factorization of the joint distribution
- Tractable conditional distributions in the correct order

In many Bayesian models, the **joint density is known only up to a normalizing constant**, and such a factorization is unavailable.

□ Gibbs Sampling

Gibbs is a Special Case of Metropolis-Hastings (acceptance probability $\gamma = 1$) in high dimension random variable $\vec{X} \in \mathbb{R}^n$.

Gibbs sampling also reduces the problem of **multivariate** sampling to the problem of **univariate** sampling.

Goal: Generate samples from $p(\vec{x}) = p(x_1, \dots, x_d)$.

Example ($d = 2$) the target distribution be a joint density $\pi(x, y)$,

Given current state (x, y) , a **general Metropolis–Hastings** proposal (x', y') is generated from a proposal density

$$q((x', y') \mid (x, y)),$$

and accepted with probability

$$\alpha = \min \left\{ 1, \frac{\pi(x', y') q((x, y) \mid (x', y'))}{\pi(x, y) q((x', y') \mid (x, y))} \right\}.$$

Gibbs Proposal for Two Variables

In the Gibbs sampler, we update **one coordinate at a time**.

Update X given $Y = y$

From current state (x, y) , propose

$$x' \sim \pi(x \mid y), \quad y' = y.$$

Thus the proposal density is

$$q((x', y') \mid (x, y)) = \pi(x' \mid y) \mathbf{1}_{\{y'=y\}}.$$

Similarly, the reverse proposal density is

$$q((x, y) \mid (x', y')) = \pi(x \mid y) \mathbf{1}_{\{y'=y\}}.$$

Acceptance Probability Calculation

Plug into the Metropolis–Hastings ratio:

$$\frac{\pi(x', y) q((x', y) | (x', y))}{\pi(x, y) q((x', y) | (x', y))} = \frac{\pi(x', y) \pi(x | y)}{\pi(x, y) \pi(x' | y)}.$$

Using the identity

$$\pi(x, y) = \pi(x | y) \pi(y),$$

we obtain

$$\frac{\pi(x', y) \pi(x | y)}{\pi(x, y) \pi(x' | y)} = \frac{\pi(x' | y) \pi(y) \pi(x | y)}{\pi(x | y) \pi(y) \pi(x' | y)} = 1.$$

Hence,

$$\alpha = 1.$$

All Gibbs proposals are accepted.

The same argument holds when updating $Y | X = x$.

Example: Bivariate Standard Normal

Let (X, Y) have a bivariate standard normal distribution with correlation ρ .

Conditional Distributions

From multivariate normal theory:

$$X \mid Y = y \sim \mathcal{N}(\rho y, 1 - \rho^2),$$

$$Y \mid X = x \sim \mathcal{N}(\rho x, 1 - \rho^2).$$

Gibbs Sampler for the Bivariate Normal

- **Initialize:** $(x_0, y_0) = (0, 0)$, set $m = 1$
- Sample

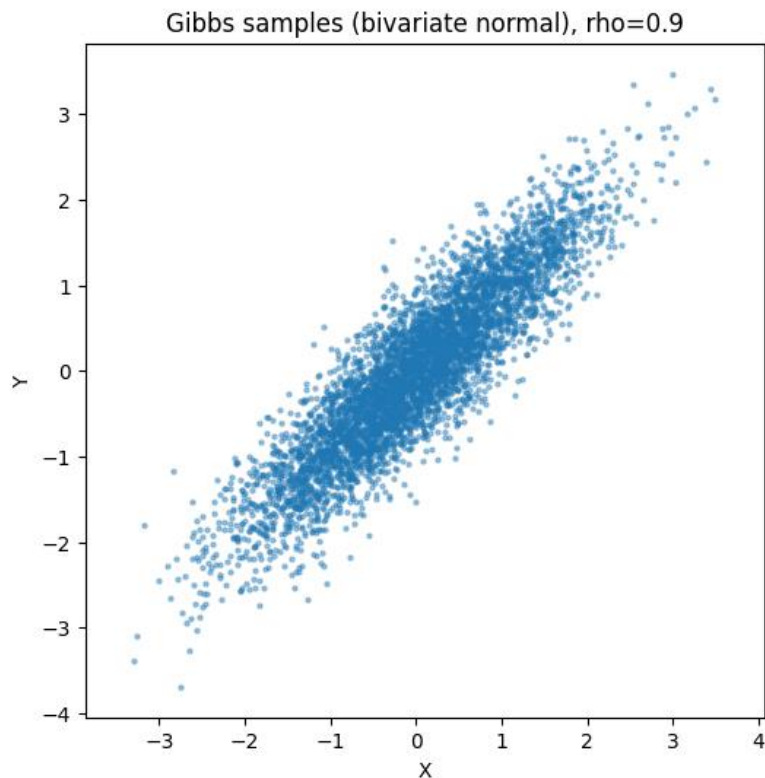
$$x_m \sim \mathcal{N}(\rho y_{m-1}, 1 - \rho^2)$$

- Sample

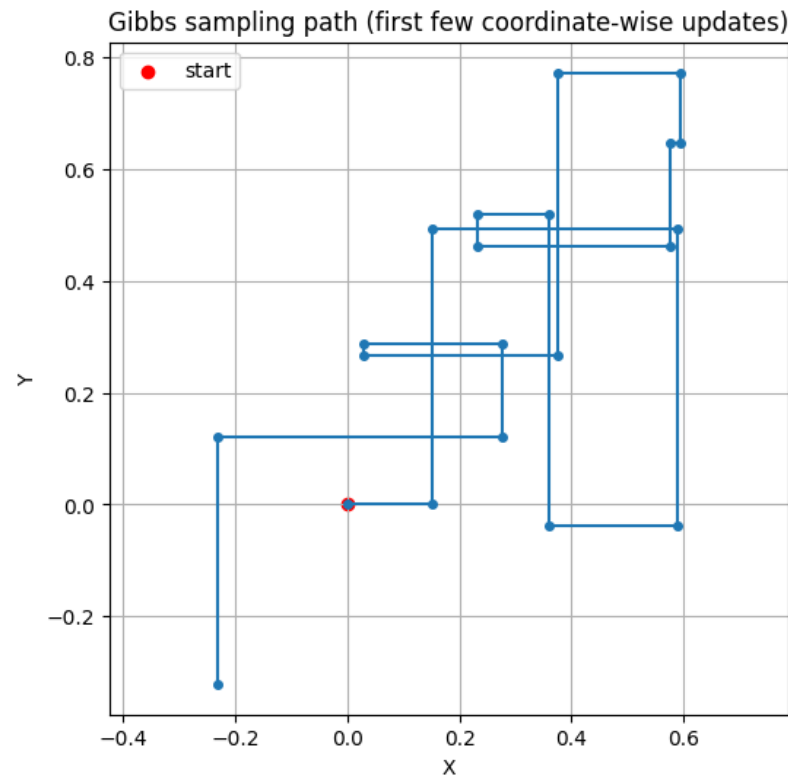
$$y_m \sim \mathcal{N}(\rho x_m, 1 - \rho^2)$$

- Set $m \leftarrow m + 1$, return to Step 2

The sequence $\{(x_m, y_m)\}$ forms a Markov chain with stationary distribution equal to the bivariate normal with correlation ρ .



Gibbs sampler for a bivariate normal



Geometrically, this produces a **zig-zag** path (horizontal then vertical moves).

<https://drive.google.com/file/d/1Vqj9BGtEgBZZH88AJ1ipuUaPGOhGVVTf/view?usp=sharing>

Gibbs Sampling (d=3)

Suppose the target distribution is a **3-dimensional joint distribution**

$$p(x_1, x_2, x_3),$$

and at iteration t we currently have the sample

$$\mathbf{x}^{(t)} = (x_1^{(t)}, x_2^{(t)}, x_3^{(t)}).$$

Our goal is to generate a new sample $\mathbf{x}^{(t+1)}$ whose distribution moves closer to the target joint distribution p .

By the **product rule**, we can write

$$p(x_1, x_2^{(t)}, x_3^{(t)}) = p(x_1 \mid x_2^{(t)}, x_3^{(t)}) p(x_2^{(t)}, x_3^{(t)}).$$

This shows that, **conditional on the current values of the other variables**, the distribution of x_1 is given by the univariate conditional density

$$p(x_1 \mid x_2^{(t)}, x_3^{(t)}).$$

Step 1: Update the First Coordinate

Draw

$$x_1^{(t+1)} \sim p(x_1 \mid x_2^{(t)}, x_3^{(t)}).$$

We assume this conditional distribution is **easy to sample from**, since it is univariate.

After this step, the state becomes

$$(x_1^{(t+1)}, x_2^{(t)}, x_3^{(t)}).$$

Step 2: Update the Second Coordinate

Now condition on the **most recent value** of x_1 and the old value of x_3 , and draw

$$x_2^{(t+1)} \sim p(x_2 \mid x_1^{(t+1)}, x_3^{(t)}).$$

The state is now

$$(x_1^{(t+1)}, x_2^{(t+1)}, x_3^{(t)}).$$

Step 3: Update the Third Coordinate

Finally, update the last coordinate using the latest values of the other two:

$$x_3^{(t+1)} \sim p(x_3 \mid x_1^{(t+1)}, x_2^{(t+1)}).$$

We obtain the next iterate

$$\mathbf{x}^{(t+1)} = (x_1^{(t+1)}, x_2^{(t+1)}, x_3^{(t+1)}).$$

Gibbs Sampling Algorithm:

- Suppose we want to obtain n samples of $x = (x_1, x_2 \dots x_d)$ from a joint distribution $p(x_1, x_2, \dots x_d)$
- Denote the i -th sample as $x^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots x_d^{(i)})$
- Gibbs Sampling:
 - 1) Randomly select initial value $x^0 = (x_1^{(0)}, x_2^{(0)}, \dots x_d^{(0)})$
 - 2) Iterate each $i = 1, 2, \dots n$:
 - Sample $x_1^{(i)} \sim p(x_1 | x_2^{(i-1)}, x_3^{(i-1)}, \dots x_d^{(i-1)})$
 - Sample $x_2^{(i)} \sim p(x_2 | x_1^{(i)}, x_3^{(i-1)}, \dots x_d^{(i-1)})$
 - ...
 - Sample $x_k^{(i)} \sim p(x_k | x_1^{(i)}, \dots, x_{k-1}^{(i)}, x_{k+1}^{(i-1)} \dots x_d^{(i-1)})$
 - ...
 - Sample $x_d^{(i)} \sim p(x_d | x_1^{(i)}, x_2^{(i)}, x_3^{(i)}, \dots x_{d-1}^{(i)})$

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Recommended textbooks

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- Probability and Random Processes - Geoffrey Grimmett and David Stirzaker
- Introduction to Probability, Statistics, and Random Processes - Hossein Poshro-Nik
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More textbooks:

- Essentials of Stochastic Processes – Richard Durrett
- Understanding Markov Chains- Nicolas Privault
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An interactive visualization of many of these algorithms in 2d, see

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http://www.stat.ucla.edu/~sczhu/Courses/UCLA/Stat_202C/Stat_202C.html (Theory)

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- <https://pat-laub.github.io/rare-events/>

- https://wires.onlinelibrary.wiley.com/doi/epdf/10.1002/wics.1314?saml_referrer
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